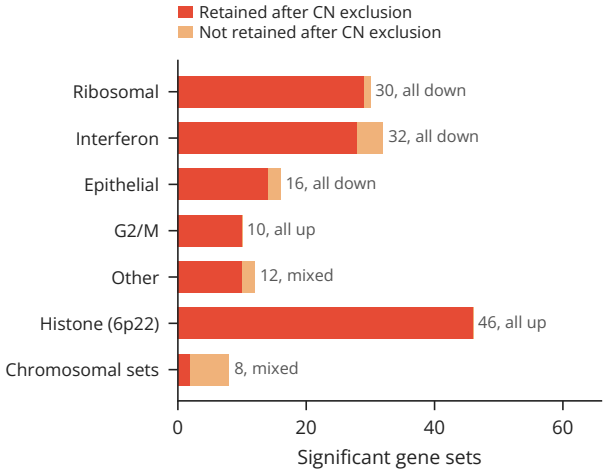
a 23,252 sets tested; 154 significant (90 lower, 64 higher)

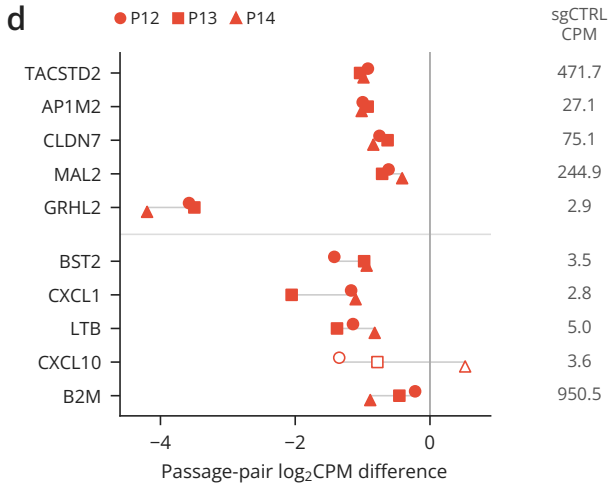
b Global FDR

	NES	Leading-edge genes	fgsea	CAMERA
Interferon and inflammatory signalling				
[H] Interferon-alpha response	*	BST2, CXCL11, RTP4	2.8e-02	1.1e-07
[H] Interferon-gamma response	*	IDO1, BST2, ST8SIA4	5.0e-03	7.6e-07
[H] Inflammatory response	*	BST2, ITGB8, TLR2	3.2e-02	2.4e-03
Epithelial-associated signatures				
[CGP] Kohn: epithelial EMT component	*	GRHL2, OVOL1, S100A14	6.8e-03	4.5e-11
[CGP] Charafe: basal vs mesenchymal, up	*	GRHL2, MACC1, PROM2	5.0e-05	1.3e-07
[CGP] Onder: CDH1 targets 2, down	*	GRHL2, IVL, CXCL1	7.2e-04	2.6e-02
[CGP] Hollern: breast-tumour EMT, down	*	GRHL2, TMPRSS13, EPS8L1	8.3e-03	1.6e-04
Ribosomal and translation-associated sets				
[MF] Structural constituent of ribosome	*	RPS9, RPS25, MRPS24	2.3e-03	4.4e-11
[R] Eukaryotic translation elongation	*	RPS9, RPS25, RPL13	5.5e-03	4.1e-21
[R] SRP-dependent protein targeting	*	RPS9, RPS25, RPL13	4.8e-03	3.2e-15
[R] rRNA processing	*	RPS9, NOP56, RPS25	3.2e-02	8.7e-07
Histone and chromatin-associated sets				
[CC] Nucleosome	*	H3-7, H1-3, H2BC4	2.9e-04	1.9e-04
[MF] Structural constituent of chromatin	*	H3-7, H1-3, H2BC4	7.2e-04	4.2e-05
[R] HDACs deacetylate histones	*	H2BC4, H2AC16, H2BC7	1.3e-03	2.6e-03
[R] PRC2 methylates histones and DNA	*	H2BC4, H2BC7, H2BC10	5.3e-03	8.8e-04
Cell-cycle and RHO-associated sets				
[R] M phase	*	TUBA3D, H2BC4, H2BC7	4.6e-02	9.6e-03
[R] RHO GTPase effectors	*	S100A9, TUBA3D, H2BC4	6.5e-03	5.5e-03
[H] G2M checkpoint		KNL1, CENPF, MYBL2	6.1e-01	4.4e-03
[H] Mitotic spindle		CDC42, OPHN1, MID1	4.7e-01	6.4e-03

c



d



* fgsea FDR < 0.05. Bold: FDR < 0.05. Leading-edge genes: three largest absolute ranking effects per set.

H, Hallmark; R, Reactome; CGP, chemical/genetic perturbations; MF/CC, GO molecular function/cellular component.